Solution Architecture - Digitize In-Store Long-Prep Ordering

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Goal

Digitize In-Store Long-Prep Orders that are placed via physical forms. Digitizing this process enables us with the following:

1. Maintain Service Department specific Menu digitally.

2. Improve Operational efficiency for effective batching of Orders during preparation.

3. Capture Order and Selections data available for reporting and personalization.

4. Multiple Service departments can onboard into this platform.

Requirements

High Level Requirements

High level requirements - This could be linked to high level requirements page from Product **but not to Jira stories**

Product Pages:
In-store bakery order digitization
In-store bakery order digitization
https://confluence.safeway.com/spaces/CDT/pages/597829096/Requirements+Areas+of+Work

Architecturally Significant Requirements

Key requirements significant from architecture standpoint that will drive key decisions in architecture:

1. AcuView Order Capture is served by new PDP XAPI which do required grouping of products

2. Order Place function will make use of Guest User. (Note: Guest User was introduced as part of Deli MTO ISM and Kiosk).

3. In Store Order's Synch "Short Code" will be used for SMS.

4. Guest User's will be retained for longer time as opposed to 7 days (per current setting).

5. Slots Target state for Digitalized InStore Orders is to use separate capacity threshold at Service Department (Deli or Bakery) level for Slots.

Architecture

Key Assumptions

Assumptions

1. Consider only Digital Stores as AcuView is available in Digital Stores.

2. iPad AcuView iPad App will depend on a Printer to print Barcode for POS to scan.

3. Customer pays In-Store Order at POS using the Barcode generated by AcuView iPad App.

4. Digital InStore Cake Order can be linked with POS Sale upon either (a) Customer provides same loyalty phone# as Digital InStore Order Intake or else (b) Associate enters POS Sales receipt# into Acuview Order during Hand-Off.
In the absence of both of them, there is an indirect way (matching Date, Product#) to link Digital Instore Order and corresponding POS Sale.

5. AcuMTO does not require to ingest Menu to serve AcuView App as NEW Ordering Web View will have indirect integration with Catalog's NEW Product-as-Menu.

6. Order Edit can be achieved by two(2) step process Cancel Order and Place New Order with Forcing Slots.

7. MLP will use open Slots by not integrating with Slots. Slots Target state for Digitalized InStore Orders is to use separate capacity threshold at Service Department (Deli or Bakery) level for Slots.

Dependencies

Key Dependencies in terms of systems/Processes both internal and external

Internal Dependencies:

External Dependencies:

Related Applications / AppCode

Applications	Components	App Code
CATALOG and XAPI	1. Product Filter API 2. PDP XAPI 3. Reverse proxy Instore DC	
AcuView	Ordering Capture View Order Prep Screens	
C & C: Cart, Checkout, Order Place, Pricing	1. Cart xAPI 2. Checkout & Pricing API & 3. Order Place API	
~~SLOTS~~	~~In-Store Order Slots & Threshold~~	
DOMS	In-Store Order	
FFM	In-Store Order Status	
ACUMTO	In-Store Order Prep	
COMMS	SMS	
UCA	Guest profile creation	
UDP Fulfilment		

Key Architecture Decisions

Key Decisions from architecture standpoint

1. In-Store Orders will utilize Omni Channel Platform and hence Orders will go through DOMS.

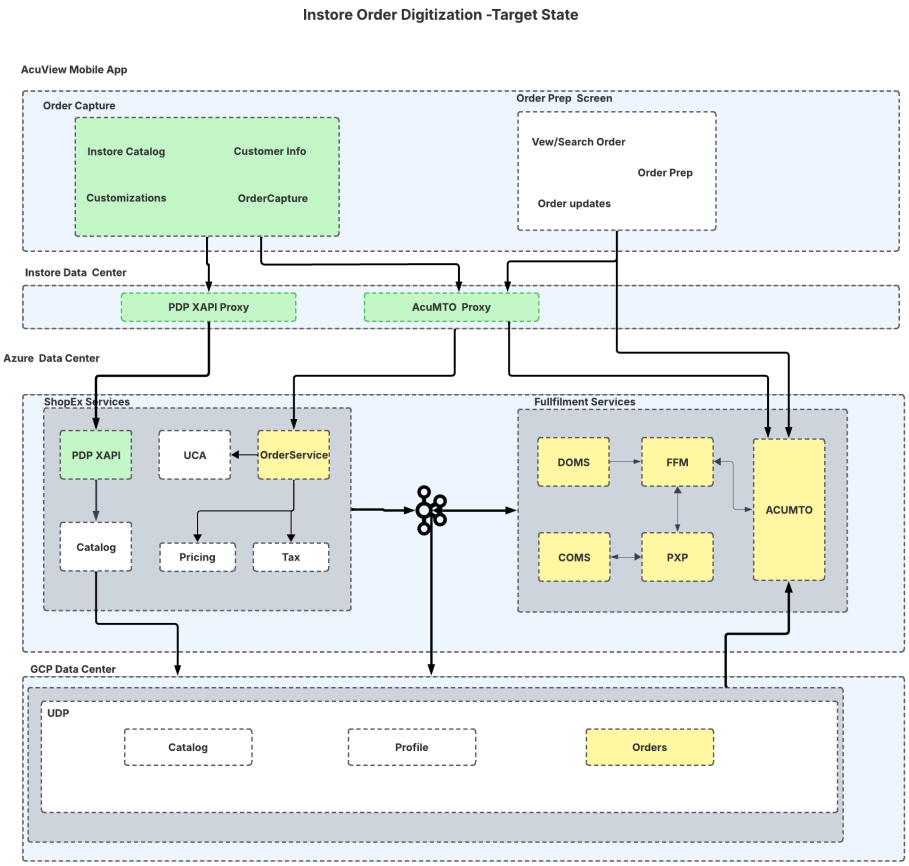
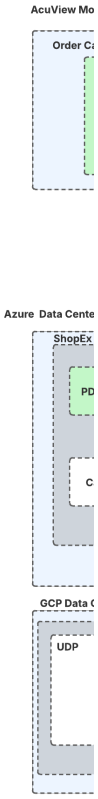
2. Digitized InStore Orders will use Guest User to capture Email and SMS details.

3. Digitized InStore Orders will use same Synch Short Code as MTO ISM and Kiosk Orders for SMS communication.

4. During Digitized InStore Order placement, Associate can either force a Slot Availability to place an Order or Slots threshold can be configured depending on the Season (example: Valentines Day, December, Games Days, Graduation etc)

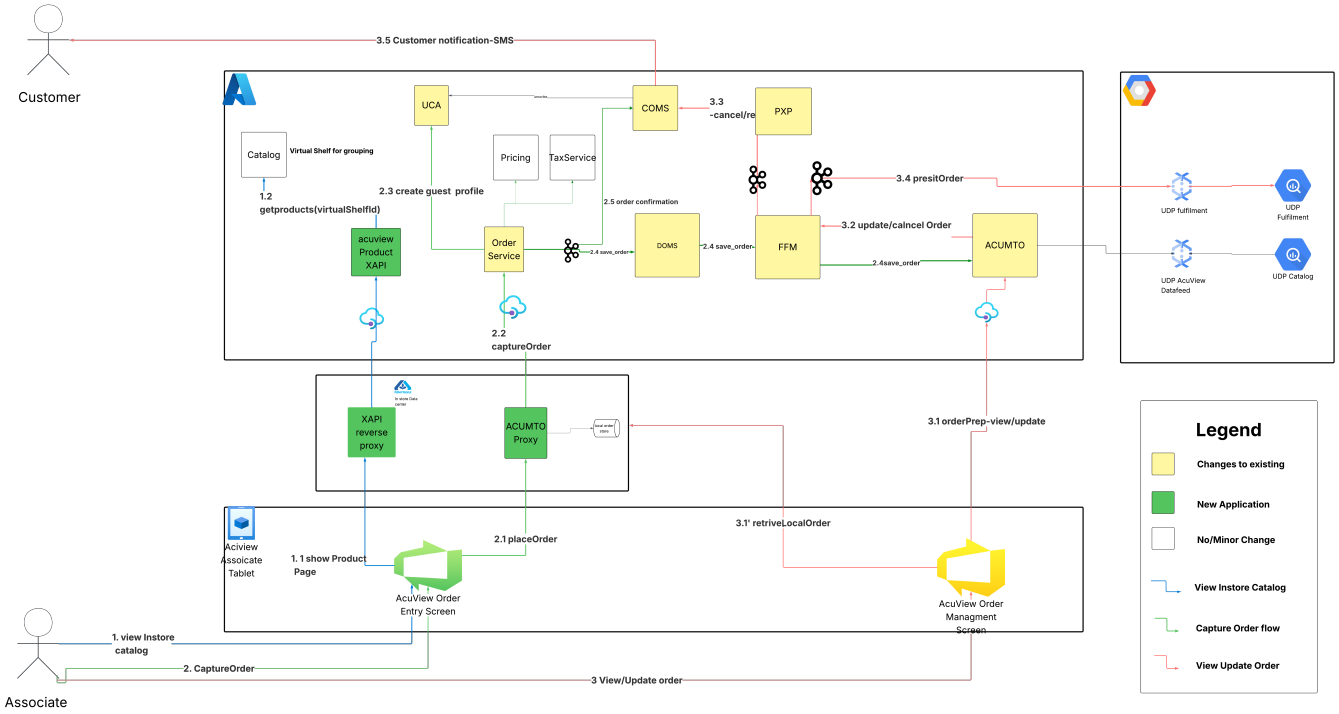
High Level Architecture

High Level Architecture Diagram and description

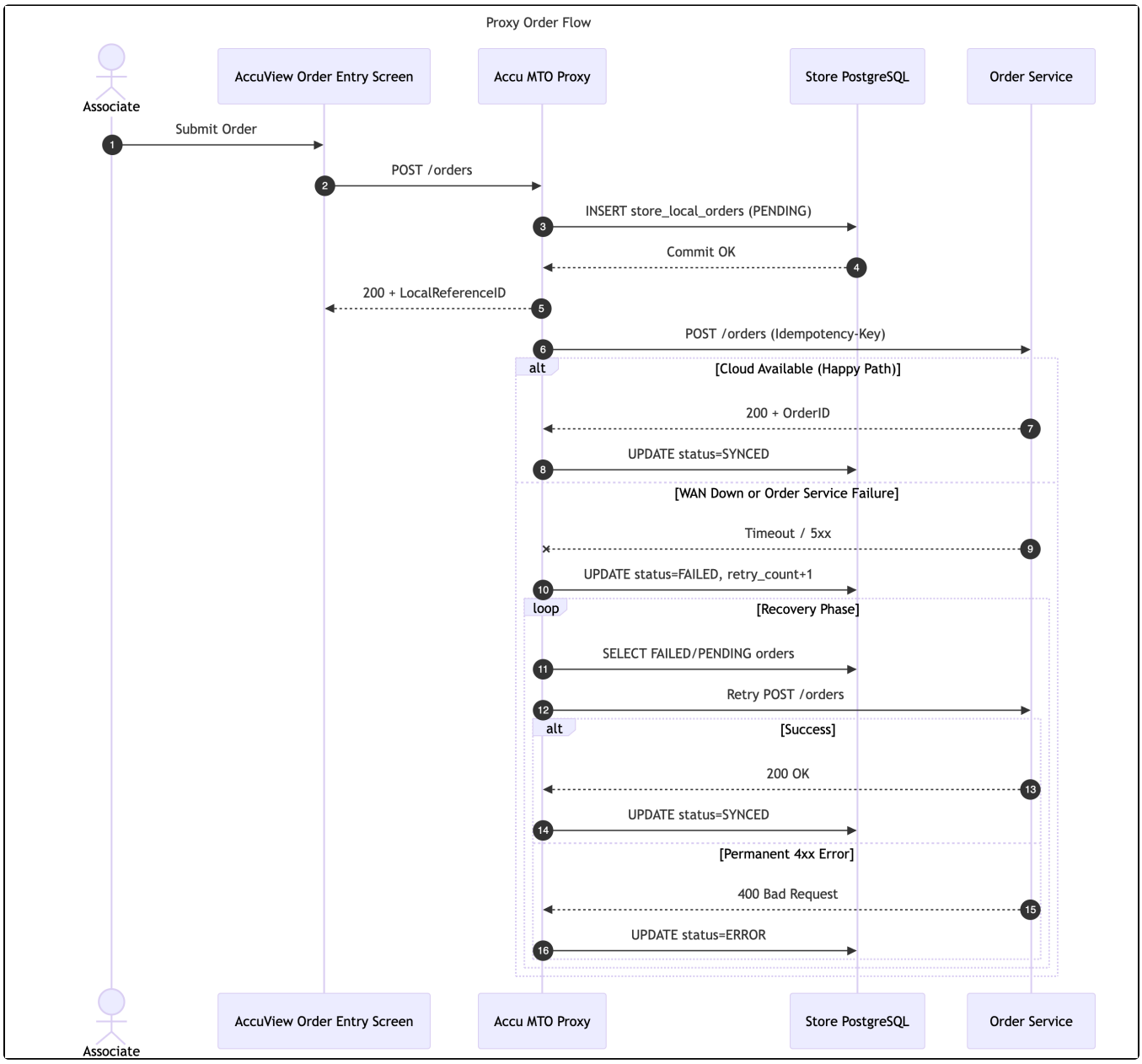
Target Architecture	Interim State
	

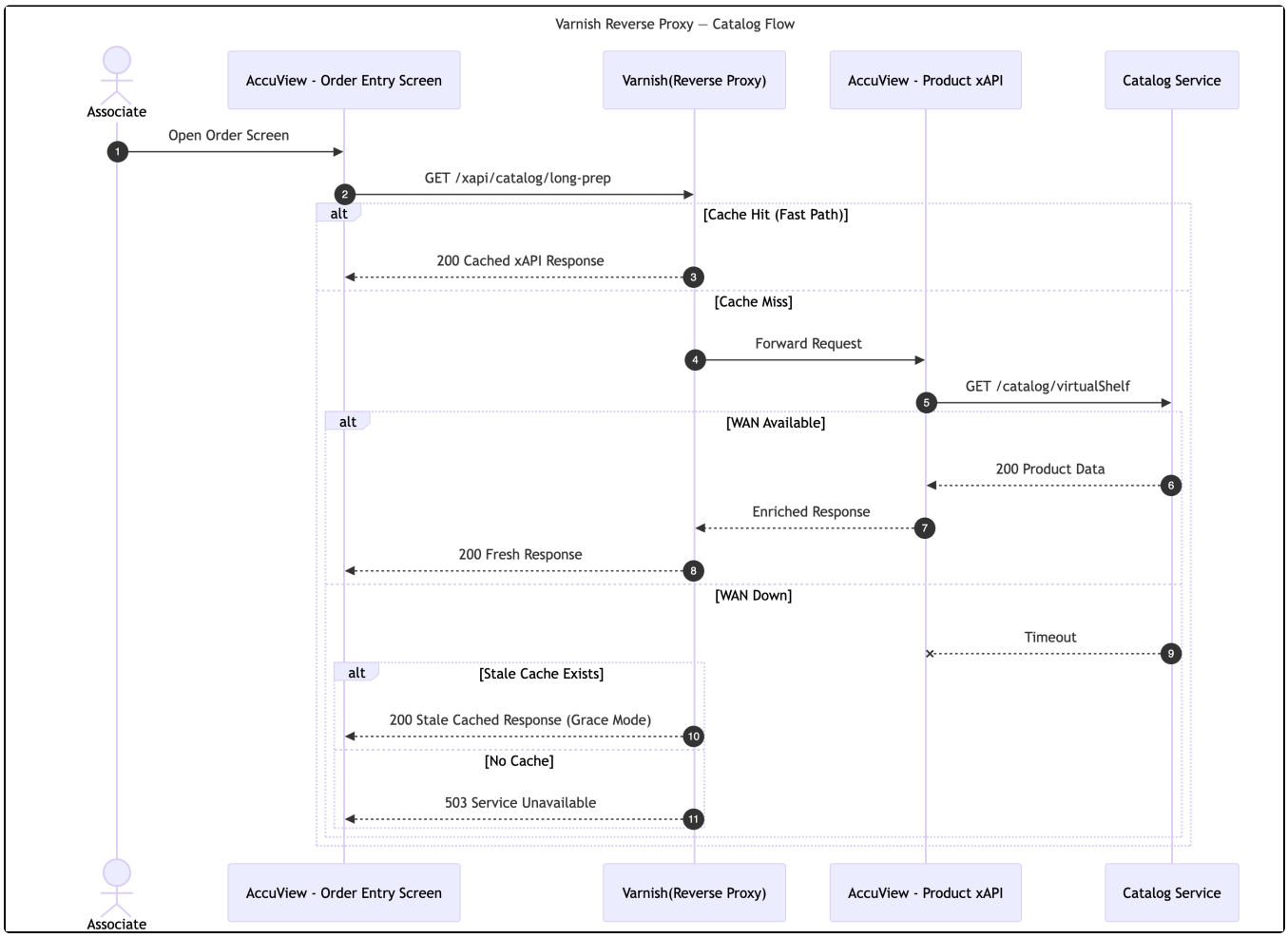
Detailed Architecture and Design

Detail Architecture with data flow diagrams, sequence diagrams of important use cases, data model, integration architecture

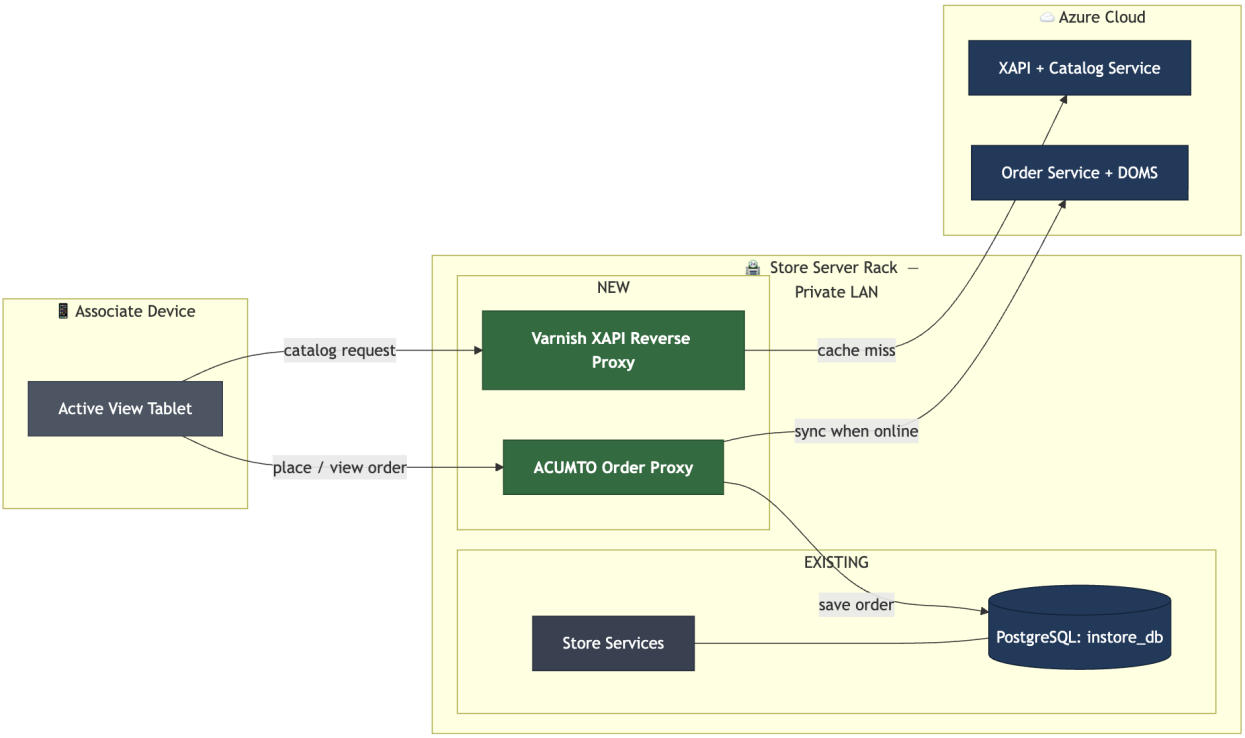


Integration Architecture


Proxy Order Flow
Associate
AccuView Order Entry Screen
Accu MTO Proxy
Store PostgreSQL
Order Service
1 Submit Order
2 POST /orders
3 INSERT store_local_orders (PENDING)
4 Commit OK
5 200 + LocalReferenceID
6 POST /orders (Idempotency-Key)
alt
[Cloud Available (Happy Path)]
7 200 + OrderID
8 UPDATE status=SYNCED
[WAN Down or Order Service Failure]
9 Timeout / 5xx
10 UPDATE status=FAILED, retry_count+1
loop
[Recovery Phase]
11 SELECT FAILED/PENDING orders
12 Retry POST /orders
alt
[Success]
13 200 OK
14 UPDATE status=SYNCED
[Permanent 4xx Error]
15 400 Bad Request
16 UPDATE status=ERROR
Associate
AccuView Order Entry Screen
Accu MTO Proxy
Store PostgreSQL
Order Service



Deployment Diagram



Data

Data Requirements

Data required for this project

As of today UDP receives In-Store Orders placed from MTO ISM and Kiosk. In-Store Orders placed from AcuView iPad also will go into same Order data tables.
UCA Guest data is being captured by UDP already.

Data Dependencies on Enterprise Systems

Dependency on Enterprise Systems for the data requirements

Data Classification

The classification of your data requirements

Data Retention Requirements

What are the data retention requirement

1. Guest User's data will be retained for longer time as opposed to 7 days (per current setting).

2. In Store Orders data retention is same as Digital Orders.

Data Privacy

Will Personal Data of any kind be collected, stored, used, accessed, shared, and/or deleted?

* Note: Personal Data includes things like name, email address, and phone number, but also includes data like IP address and machine signature as well. This data could be about customers, patients, associates, vendors, or other individuals.

As of today, In-Store Orders placed from MTO ISM and Kiosk are capturing Email and Phone#. PII and TCPA requirements are already handled at UCA.

Legal Considerations (If required)

Are there any legal considerations / requirements that need to be addressed – e.g., trade secret / intellectual property rights, marketing compliance, new consumer offering, AI use case review, etc.?

Legal team confirmation about consent taken by associate and linking order to existing loyalty

In-store order digitizat...n - legal guidelines.eml

RE_ In-store order dig...tion - SMS consent.eml

Re: Request for Privac...tization (FY26 Q1).pdf

Re: Hosting strategy fo...zation applications.pdf

Re_ Request for Privac...tization (FY26 Q1).eml

Risk

Open Questions

S. no	Question /Concern	Resolution	State
1	We should not be using BPN for Menu Items, this will complicate the Indexing. Menu Items should either be aligned to bundle concept of Meals or Digital aisle.	We would be using virtual shelf grouping which is inbuilt feature catalog	Resolved
2	In store applications should be have a resilience in their architecture that they can operate with or without cloud services. Designing the application only on cloud services can be an option but not a preferred one architecturally.	New PDP XAPI reverse proxy server in store datacenter which would serving catalog data connectivity issues	Resolved
3	The store should be able to serve the customer i.e. take orders, work on orders with limited to no connectivity, and should sync with cloud services whenever required.	New ACUMTO proxy server which would persist order in Instore datacenter when cloud connectivity issues	Resolved
4	Why we are not offering Delivery or DUG for instore orders like Ecommernce	Response from Product-"Currently not in our roadmap. If customer wants to use DUG /Delivery, the associates can nudge them to place their order on ecom App/Web to get DUG /delivery options" Saying that technical we are use same tech stack and similar order workflow as Ecommerce orders. In future , this requirement can be incremental developed	
5	Since guest profile is going to be more than 30 days . we the review from CCPA team		
6	Why cannot associate the order with existing loyalty instead of creating new guest profile	To associate loyalty number in the background we need additional consent and flows.We need more vetting from legal and product for doing this. Currently it not priority because when customer checkout at POS he/she will be able to use loyalty and get the rewards or deals	
7	What is timeline for interim state		
8	Should we think about passing over ACI tablet order entry form to customer so that they can review the order, give consent to storing order details (privacy policy)	Response from product : " No, the plan is to have the associates use the tablet and collect customer's contact info, order details, and consent. This process of associate taking customer's consent is already reviewed with legal and privacy. There will be an order confirmation screen which associate can review with customer and also customer will get order confirmation text message. These are ACI tablets which has other apps/interfaces. Since the tablet will also have AcuView OrderPrep as an example, with other customer order details, if we hand the ACI tablet to the customer, they may accidentally open other tabs/windows where they could be able to see other customer's order details. That is a concern that we have on handing over ACI tablets to customer. "	

Approvals

Approver	Approval Status	Comments	Additional Details	
Shekhar Kale	Approved 16 Apr 2026			
Nitin Saksena				
Avijeet Banerjee				